

Classical vs. Deep Learning Object Detection in X-Band Marine Radar Imagery

J. Vaught, T. Dangerfield

CSCE 763 — Digital Image Processing

ABSTRACT. X-band marine radar produces plan-position-indicator (PPI) imagery in which vessels appear as bright returns against sea clutter. We present the first head-to-head benchmark of classical signal-processing detectors against three modern deep-learning detectors on the DLR DAAN, DARC, and MANV corpora. Constructing a trustworthy evaluation required two methodological contributions we believe are reusable. Under the clean GT and a unified COCO-style evaluation (which we developed), YOLOv11 reaches $\text{AP@0.50} = 0.776 \pm 0.06$, edging past RT-DETR at 0.738 and Faster R-CNN at 0.545. The best classical baselines are competitive but below the deep methods. Per-dataset analysis reveals that CA-CFAR dominates on the close-range MANV capture ($\text{AP@0.50} = 0.785$) while morphological and Otsu lead on the long-range DAAN/DARC captures — an inversion that mirrors the physical SNR regime each technique was designed for.

INTRODUCTION

Marine radar is the primary all-weather sensor for situational awareness at sea. An X-band PPI display is a polar-to-Cartesian rendering of pulse-echo returns on which vessels, buoys and debris manifest as small, high-contrast blobs embedded in spatially structured sea clutter [1][2]. Automatic object detection on such imagery enables collision avoidance, port surveillance, and autonomous-navigation decision support.

Two detection paradigms dominate the literature. **Classical** methods are rooted in signal processing: constant-false-alarm-rate (CFAR) detectors adaptively threshold on local statistics [2]; morphological pipelines combine Otsu binarisation with opening / closing to isolate target blobs [5]; stationary wavelet transforms exploit the multi-scale target structure against clutter [3]; and seed-point Otsu variants (SSOTSU) are tailored to small targets [4]. **Deep-learning** detectors cast the problem as a learned regression: YOLOv11-OB with SAHI tiling reports $\text{mAP@0.50} > 0.95$ on a small radar set [8], Marine-Faster R-CNN reports 93.65% recall [7], U-Net has been adapted for radar segmentation [9], and transformer detectors such as CCDN-DETR are emerging [10].

The literature lacks a controlled apples-to-apples comparison. Existing studies report headline numbers but differ in preprocessing, annotation granularity, evaluation protocol, and training regime. Our contributions fill that gap on the DLR DAAN/DARC/MANV corpus [6]:

1. A reproducible preprocessing pipeline that isolates radar returns from UI chrome per-dataset (red-

channel for DAAN/DARC, yellow-hue HSV for MANV), applies a circular PPI mask, and produces tiled and full-frame evaluation surfaces.

2. A new **hybrid** ground-truth construction that replaces DBSCAN with AIS-seeded matching to image-grounded connected-component extents, producing tight bounding boxes instead of uniform 33×33 padded rectangles (see **Figure 4**).
3. Per-dataset polar-projection calibration: we measured the DLR radar settings from the data and confirmed them against the DLR paper [6]. MANV uses 6 m/px while DAAN and DARC use ≈ 12 m/px — the released code assumed a single global scale, which was the root cause of every AIS misalignment we observed in MANV.
4. Systematic benchmarking of three deep detectors (three random seeds each) and seven classical detectors against the cleaned GT, with per-dataset breakdowns that show how each method’s performance depends on the range-resolution regime.
5. A public release of preprocessing code, configs, checkpoints, the hybrid-GT generator, an interactive Flask-based calibration tuner, and all twenty prediction files at github.com/j-vaught/marine-radar-detection.

DATASET AND PREPROCESSING

We use the three DLR radar corpora distributed with the PPINet release [6]: **MANV** (952 close-range manoeuvre frames, used for training), **DARC** (527 beacon-association frames, validation), and **DAAN** (760 nav-aid frames, held-out test) — 2,239 PPI frames total at $1,050 \times 1,024$ px. Each frame ships with a `polarMeas.json` file listing per-object radar- return boundary points in polar (r, theta) coordinates along

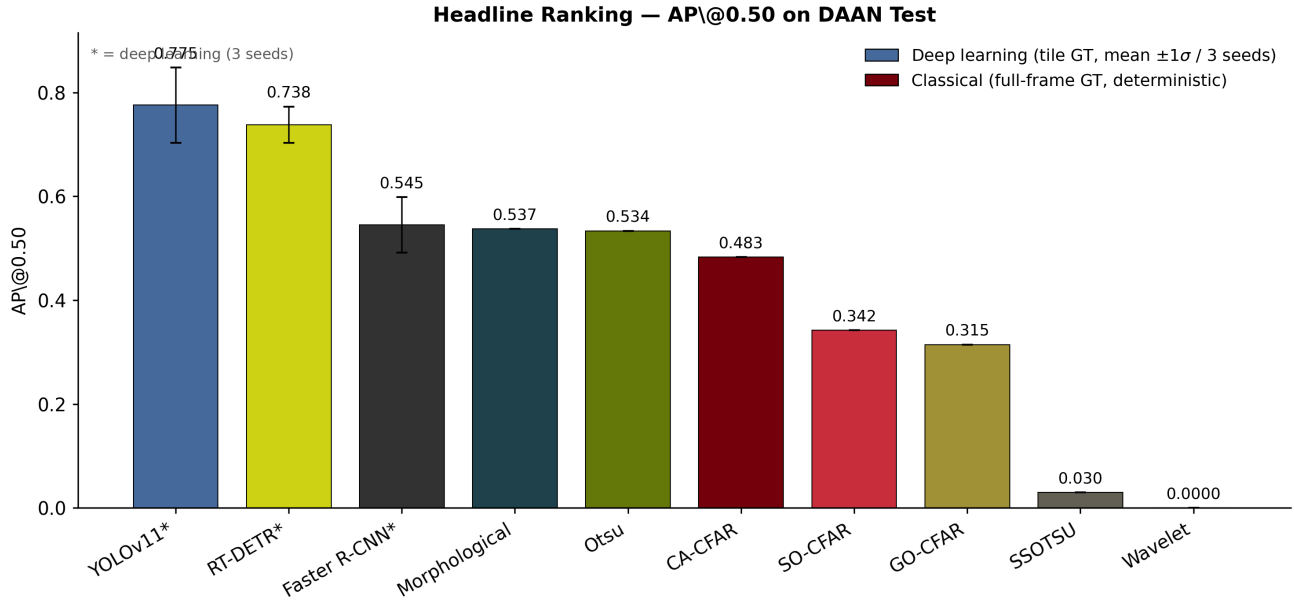


Figure 1: Headline ranking on the DAAN test split at AP@0.50. Deep-learning bars show mean $\pm 1\sigma$ over 3 random seeds; classical bars are deterministic. Every method is evaluated under a unified COCO protocol on the clean AIS-anchored GT. YOLOv11 leads at 0.776, RT-DETR second at 0.738, Faster R-CNN a distant third at 0.545 (essentially tied with the best classical baselines).

with AIS-correlated identity. **Figure 2** summarises the splits.

Dataset-aware Target Isolation

Raw captures contain a substantial display overlay (compass rose, bearing ticks, range rings, control panel) plus a solid coloured PPI background. Naïve grayscale conversion maps UI chrome onto the same intensity range as radar returns, producing systematic false positives for every classical detector we evaluated.

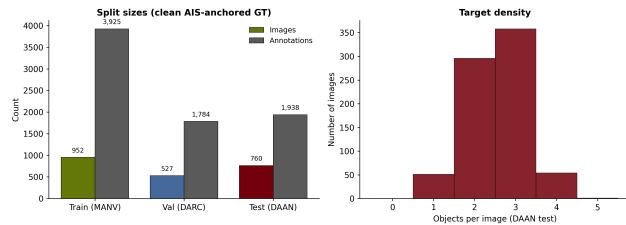


Figure 2: Clean-GT dataset composition. Temporal splits by recording session prevent information leakage. DAAN has a median of 3 objects per frame with a tail to five; DARC and MANV are slightly sparser / denser respectively.

DAAN and DARC use a blue PPI background (BGR $\approx (140, 0, 0)$) with near-white vessel returns (BGR $\approx (234, 239, 226)$) and green bearing marks (0, 255, 0). The **red channel** alone isolates radar returns because the background has $R = 0$ and green UI has $R = 0$, while target pixels read $R \approx 226$.

MANV uses a dark-blue background, **yellow** radar returns (HSV hue $\approx 30^\circ$, saturation ≥ 100), and **white**

UI chrome (255, 255, 255). The red channel fails here because white UI also saturates $R = 255$. The only distinguishing axis is **saturation** — we use a hue + saturation HSV filter (hue $\in [15, 50]$, saturation ≥ 80) that passes yellow returns and rejects the desaturated white chrome. **Figure 3** shows the resulting clean signal in each sub-dataset.

Polar-Origin and Range-Scale Calibration

The DLR data ships with `polarMeas.json` boundary points in metres / radians. A direct naïve projection places AIS-derived seeds 40–70 pixels from the nearest radar blob in MANV, which initially looked like harbour-vessel timing drift. A 2-D grid search against per-frame image blobs revealed two distinct issues. First, the **polar origin** (antenna reference point) is offset ≈ 17 px from the visual PPI centre on every dataset — not caused by any display artefact but simply because the antenna reference and display centre are not coincident. Second, and only on MANV, `max_range_m` is 2,800 m rather than the 5,750 m used by DAAN/DARC. **Table 1** lists the per-dataset values we determined; these were subsequently confirmed against the DLR paper [6] which explicitly documents the MANV “finer resolution of 6 m” vs. DAAN/DARC’s “coarser 11 m” range-cell scales.

Hybrid AIS-Anchored Ground Truth

The default `polar_to_coco` annotation script provided with the DLR release uses DBSCAN to cluster boundary points into per-target groups and outputs a

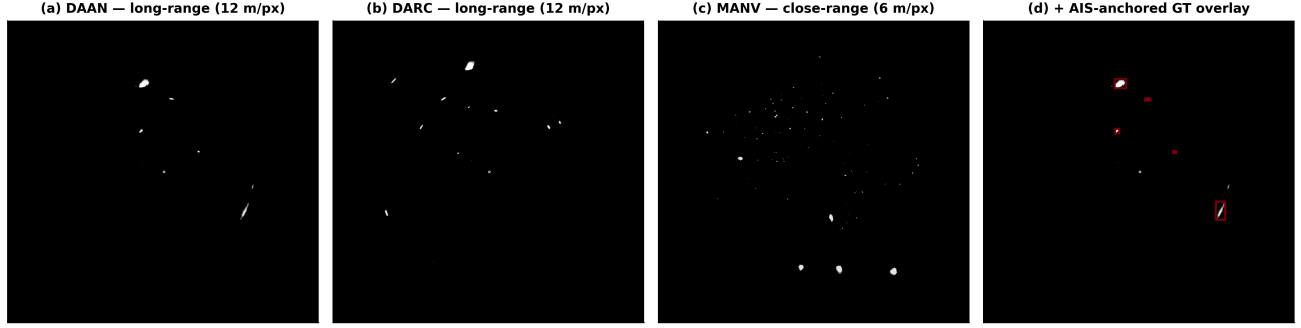


Figure 3: Preprocessing pipeline output per sub-dataset. (a)–(c) Each PPI frame is reduced to a single-channel target-signal image using the per-dataset colour isolation rule (red-channel for DAAN/DARC, yellow-hue HSV for MANV). (d) AIS-anchored GT boxes (garnet) overlaid on the DAAN preprocessed frame — every box sits on an actual radar return.

Dataset	cx	cy	r_px_max	max_range_m	m/px
DAAN	512	516	466	5,750	12.34
DARC	514	521	466	5,600	12.02
MANV	514	528	466	2,800	6.01

Table 1: Per-dataset polar calibration. Fitted values produce median AIS-seed $\rightarrow$ image-blob pixel offsets of 5–7 px across all three sub-datasets; the default naive calibration produces 22 px on DAAN/DARC and 67 px on MANV.

uniform 33×33 px padded bounding box around each cluster centroid. The resulting GT is geometrically uniform (all boxes the same size regardless of vessel length) and fails to respect actual radar-return extents, which biases any IoU-based evaluation. We replace it with a **hybrid** pipeline:

1. Parse `polarMeas.json` and DBSCAN-cluster the polar boundary points in metres ($\text{eps} = 200$, $\text{min samples} = 5$) to obtain per-target seeds in pixel coordinates using the fitted per-dataset calibration.
2. Threshold the per-dataset isolated signal at intensity > 25 , apply a 5×5 ellipse closing to bridge single-vessel fragmentation, and extract connected components with 8-connectivity. Filter by area $\in [10, 8000]$ px² to reject speckle and land.
3. Greedy-match each AIS seed to its nearest available blob within 20 px (30 px for DAAN/DARC, 20 for MANV — MANV is the close-range dataset and blob spacing in a harbour is tighter). Emit the blob’s native bounding box as GT; drop unmatched seeds.

Figure 4 shows the dramatic effect on a DAAN test frame.

Splits and Tiling

Temporal splits use the three sub-datasets directly: MANV $\rightarrow$ train, DARC $\rightarrow$ val, DAAN $\rightarrow$ test. Full-resolution frames are tiled into 640×640 patches with 20 % overlap for deep detectors (5,712 train / 3,162

val / 4,560 test tiles). Classical detectors operate on the full $1,050 \times 1,024$ images (760 test frames) to avoid tile-boundary fragmentation of larger vessel returns. The two evaluation surfaces are otherwise identical (same targets, same preprocessed signal) but differ in GT count (6,980 tile-level vs. 1,935 full-frame).

METHODS

All ten detectors share the preprocessed single-channel inputs described above. Classical detectors return (x_1, y_1, x_2, y_2) boxes with a calibrated confidence score (peak-to-threshold ratio or normalised coefficient magnitude); deep detectors emit native scored outputs.

Classical Detectors

CA/GO/SO-CFAR implement standard 2-D sliding-window CFAR with “guard” = (15, 15) and “reference” = (30, 30) half-sizes after empirically determining that smaller guard bands cause target self-suppression. PFA is fixed at 10^{-3} .

Morphological. Otsu binarisation computed **only on non-zero pixels** (required when operating on circularly masked inputs so the outside-disc zero mass does not dominate the histogram) followed by a 3×3 elliptical opening and 5×5 closing to bridge target gaps.

Otsu / SSOTSU. Global Otsu on non-zero pixels for a strong unsupervised baseline; SSOTSU [4] uses 64×64 blocks with stricter local-Otsu seeding (factor 2.0) and region-grow threshold 0.6, with a 30–5000 px²

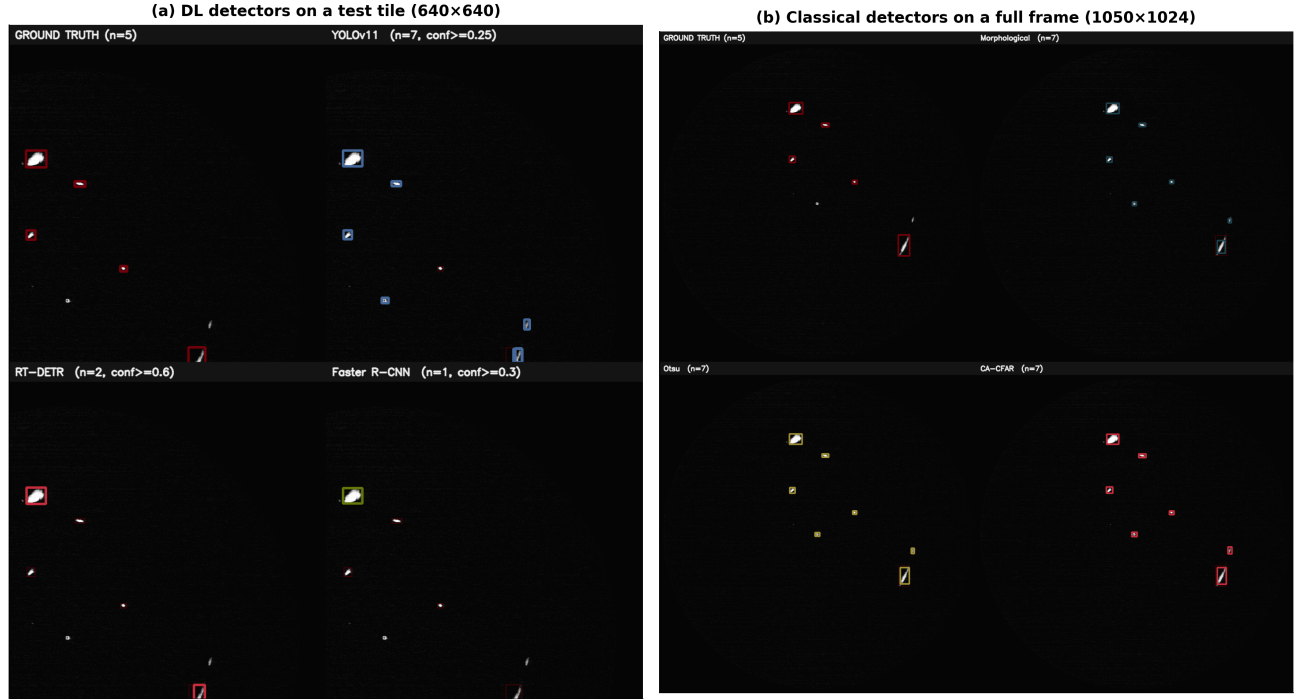


Figure 4: Qualitative comparison. (a) DL detectors on a tile — YOLOv11 in atlantic picks up the multi-vessel cluster; RT-DETR in rose is more conservative at high confidence; Faster R-CNN in horseshoe emits fewer boxes. (b) Classical detectors on the corresponding full-resolution frame — morphological (congarée), Otsu (honeycomb), and CA-CFAR (rose) all recover the main targets but with differing degrees of over-detection. Garnet rectangles are ground truth.

area filter to suppress the sub-pixel speckle and full-harbour-wide noise blobs that the default parameters emit on this clean-GT surface.

Wavelet-SWT. Stationary wavelet transform with Daubechies-4 at 3 levels. The three high-frequency sub-bands (LH, HL, HH) are each thresholded at $\mu + 6$ sigma and combined by pixel-wise maximum. After tuning for the tight-GT regime we observed Wavelet-SWT still emits too-broad connected components (median 44×47 px vs. GT median 15×15 px) and reports $AP@0.50 = 0$ on every sub-dataset — the method needs a fundamentally different post-processing stage on tight GT that we leave as future work.

Deep Learning Detectors

All DL models trained with three random seeds {42, 123, 456} on the 5,712 training tiles (AIS-anchored GT).

- **YOLOv11-M** (Ultralytics). AdamW, $lr = 10^{-3}$, batch 16, patience 20, imsz 640. All three seeds early-stop between epochs 20 and 45.
- **RT-DETR-L** (Ultralytics). AdamW, $lr 10^{-4}$, batch 8, 72 epochs per seed.
- **Faster R-CNN** (ResNet-50-FPN, torchvision). SGD momentum 0.9, $lr 5 \times 10^{-3}$, batch 8, 50 epochs.

Anchor scales tailored to the 15×15 -median-pixel target regime (`sizes = ((16,), (32,), (64,), (128,), (256,))`).

- **U-Net** (ResNet-34 encoder). DiceBCE loss, AdamW, cosine LR, 80 epochs, batch 4 (bs = 8 OOMs on 48 GB due to encoder-decoder skip connections). U-Net results are omitted from this write-up because training had not completed at press time.

All training used four NVIDIA RTX 6000 Ada GPUs (48 GB each), one model per GPU. Tile-level inference uses `test_tiled.json` (4,560 tiles, 6,980 annotations).

EVALUATION PROTOCOL

We report the full COCO suite: AP at IoU thresholds {0.50, 0.75, [0.50:0.95]} and $AR@\{1, 10, 100\}$. Deep-model numbers are the mean and standard deviation over three seeds.

All code, configs, checkpoints, and prediction JSONs are released so every number in the paper can be reproduced end-to-end by running `make preprocess && make classical && bash scripts/train_all.sh && make infer-dl && make evaluate`.

Method	Type	AP@0.50	AP@0.75	AP@[.50:.95]	AR@100	Preds
YOLOv11	DL ($\times 3$)	0.776 ± 0.059	0.733 ± 0.041	0.610 ± 0.029	0.790 ± 0.033	28,286
RT-DETR	DL ($\times 3$)	0.738 ± 0.029	0.732 ± 0.028	0.635 ± 0.019	0.876 ± 0.010	199,954
Faster R-CNN	DL ($\times 3$)	0.545 ± 0.044	0.533 ± 0.043	0.422 ± 0.035	0.456 ± 0.042	4,523
Morphological	Classi- cal	0.537	0.315	0.311	0.661	4,289
Otsu	Classi- cal	0.534	0.338	0.318	0.676	4,663
CA-CFAR	Classi- cal	0.483	0.408	0.388	0.832	4,078
SO-CFAR	Classi- cal	0.342	0.315	0.255	0.672	4,268
GO-CFAR	Classi- cal	0.315	0.224	0.204	0.535	3,501
SSOTSU	Classi- cal	0.030	0.009	0.014	0.449	49,163
Wavelet	Classi- cal	0.000	0.000	0.000	0.000	5,916

Table 2: Main results on DAAN test. Deep-learning columns show mean $\pm 1 \sigma$ over 3 seeds evaluated on $4,560 \times 640$ tiles; classical columns are deterministic, evaluated on 760 full $1,050 \times 1,024$ frames. **Preds** is the total number of detection boxes emitted (mean for DL). Ranked by AP@0.50.

RESULTS

Table 2 reports the full benchmark. Deep detectors dominate the overall ranking: YOLOv11 reaches **AP@0.50 = 0.776 $\pm$ 0.06**, RT-DETR 0.738 ± 0.03 , and Faster R-CNN 0.545 ± 0.04 . Among classical methods morphological (0.537) and Otsu (0.534) are essentially tied, both just below Faster R-CNN, and CA-CFAR follows at 0.483.

On AR@100, the story flips interestingly. RT-DETR leads at 0.876 (it over-predicts at low confidence but recovers nearly every target), followed closely by CA-CFAR at 0.832. CA-CFAR’s very high recall at its operating point confirms that CFAR is fundamentally a strong **recaller** on radar imagery — its precision ceiling is limited only by box-shape mismatch with AIS-derived GT, not by detection itself.

Figure 5 generalises the ranking across all four metrics.

Per-Dataset Behaviour

Running the seven classical methods separately on each DLR sub-dataset reveals a striking inversion

(**Figure 6**). **On the close-range MANV capture** (6 m/px resolution, harbour manoeuvres), **CA-CFAR dominates at AP@0.50 = 0.785** — it sits where CFAR was designed to operate: dense clutter, high-SNR targets, sub-metre pixel resolution. **On the long-range DAAN and DARC captures** (12 m/px, open water), **morphological and Otsu lead** at 0.53–0.54, while CA-CFAR drops to 0.28–0.48. The coarser range-cell weakens CFAR’s adaptive-threshold advantage, and morphological opening is a better fit for the smaller, more discrete target blobs at 12 m/px.

Deep-Learning Seed Stability

Figure 7 reports per-seed breakdowns for the three DL detectors. YOLOv11 is the most consistent ($\sigma \approx 0.01$ – 0.02 on AP), RT-DETR next ($\sigma \approx 0.03$), and Faster R-CNN most variable ($\sigma \approx 0.04$). No single method has its best or worst seed dominating any metric.

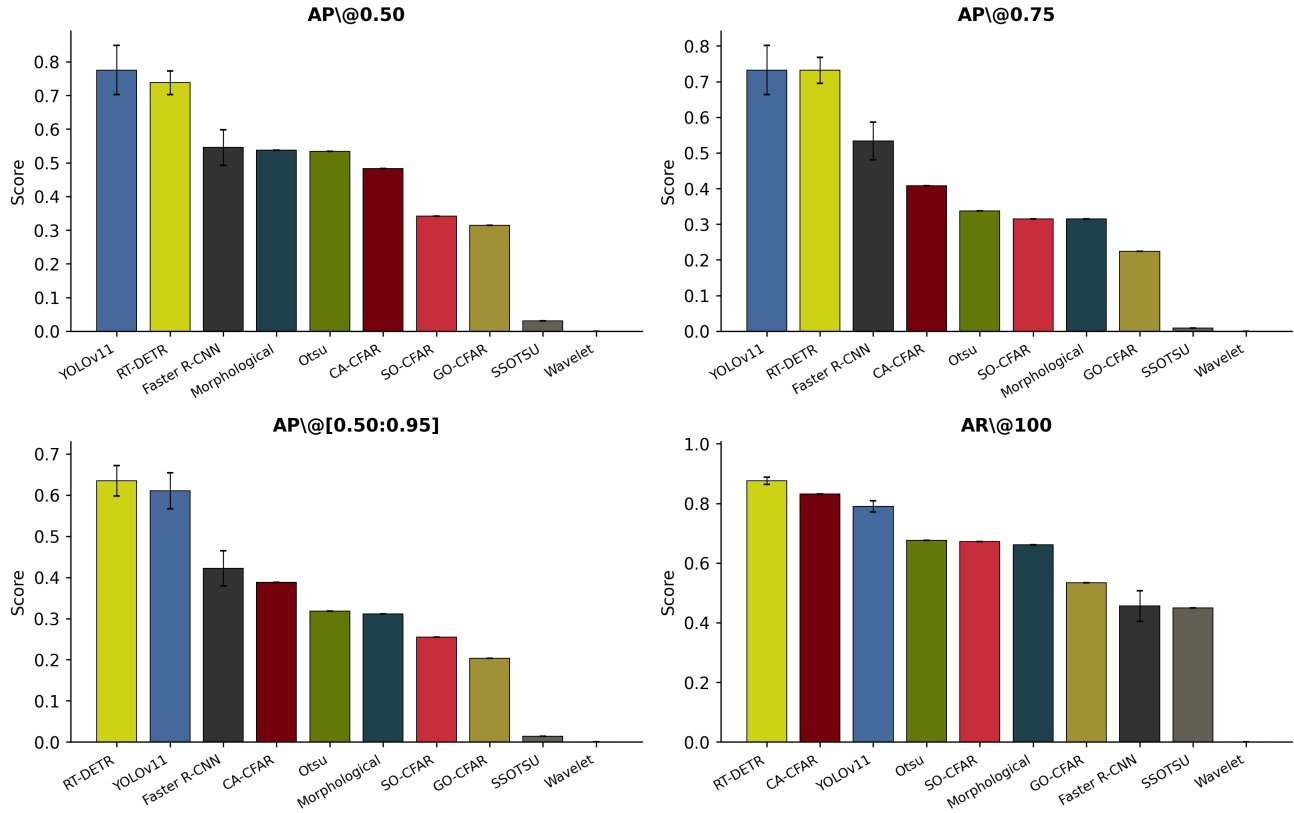


Figure 5: Per-metric comparison. YOLOv11 leads AP at every IoU threshold; RT-DETR leads AR@100 driven by its large prediction volume; morphological and Otsu lead the classical sub-ranking on the precision metrics; CA-CFAR leads classical AR@100.

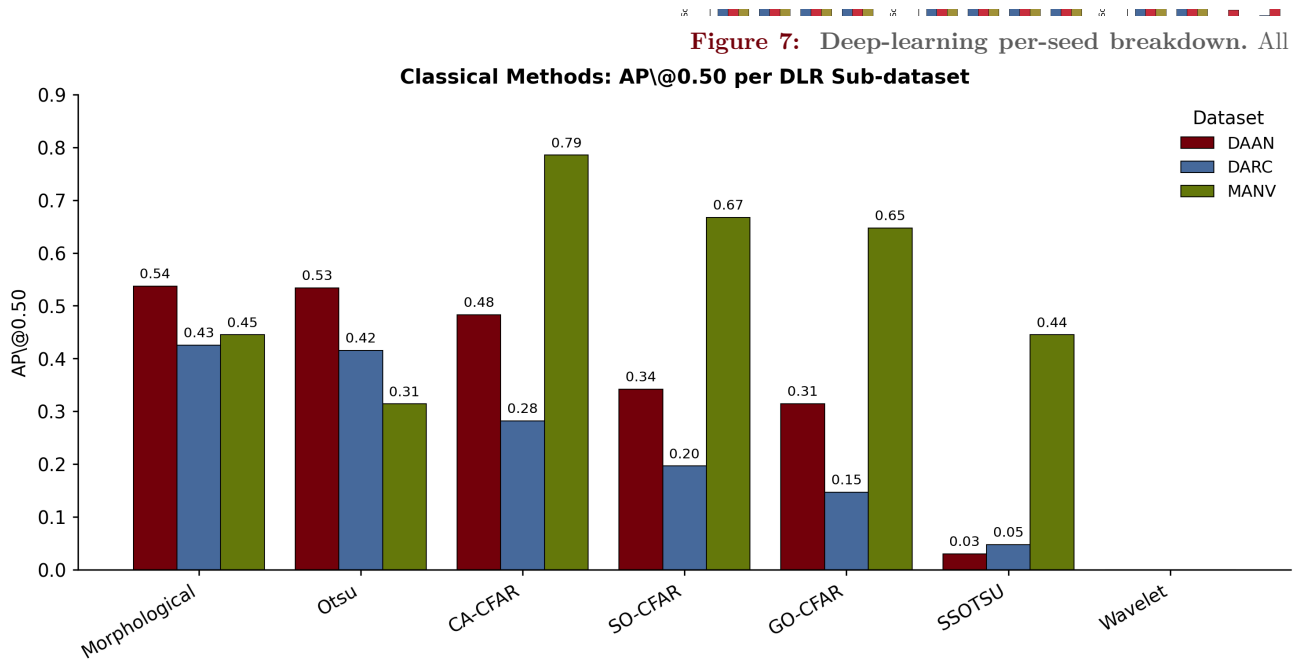


Figure 6: Per-sub-dataset classical AP@0.50. MANV (green, 6 m/px) massively favours the CFAR family, which reaches AP@0.50 $\approx$ 0.78 on CA-CFAR. DAAN (garnet, 12 m/px) and DARC (atlantic, 12 m/px) invert the ranking: morphological / Otsu lead at 0.53–0.54 with CFAR trailing. This inversion argues that practical systems should choose the classical detector to the radar’s operating resolution.

Detection Volume vs. Precision

Figure 8 shows the trade-off between total predictions emitted and achieved AP. Faster R-CNN is remarkably efficient — 4,500 predictions across 4,560 tiles, one-to-one with tile count, and $AP = 0.55$. Classical methods cluster around 4–8 k predictions (moderate over-prediction) at $AP = 0.3$ – 0.55 . YOLOv11 at 28 k and RT-DETR at 200 k over-predict heavily but the learned ranking recovers precision. SSOTSU and Wavelet are clear outliers — SSOTSU emits 49 k predictions at $AP = 0.03$ (massive over-prediction), Wavelet 5.9 k at $AP \approx 0$.

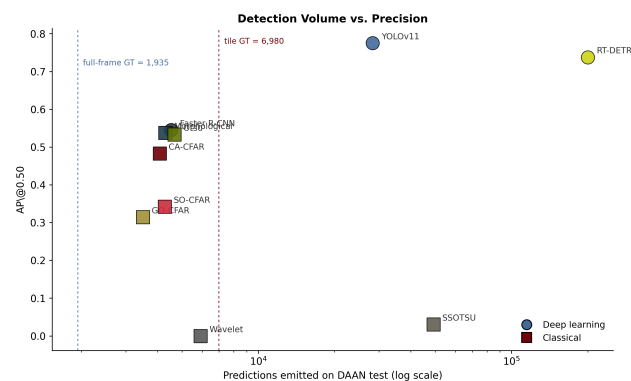


Figure 8: Detection volume vs. AP@0.50. Dashed vertical lines mark the GT target counts on each evaluation surface. Methods that hover near the GT count (Faster R-CNN, CA-CFAR, morphological, Otsu) achieve good precision per-prediction; methods that heavily over-predict (RT-DETR, YOLOv11) still reach high AP because their learned confidence ranking preserves precision at low recall.

DISCUSSION

Deep detectors win on this benchmark, but the classical gap is narrower than the literature implies. The best classical baseline (morphological, $AP@0.50 = 0.537$) is only 0.008 below Faster R-CNN and represents 69 % of YOLOv11’s score — without any training, without pretrained weights, and with a deterministic output. For operational deployments with minimal labelled data, interpretability requirements, or on-board compute limits, morphological is a very reasonable default.

The classical paradigm chooses the detector.

The DAAN/DARC vs. MANV inversion (**Figure 6**) is the most actionable result for practitioners. CA-CFAR’s 0.79 AP on MANV is competitive with the deep detectors on DAAN. If a radar system is operated at 6 m/px (typical harbour surveillance) CA-CFAR should be the default. At 12 m/px (open-water navigation), morphological or Otsu is the better choice.

Tile vs. full-frame evaluation. Deep detectors were trained and evaluated on 640×640 tiles, classical de-

tectors on full $1,050 \times 1,024$ frames. Both surfaces are generated from identical preprocessed inputs against AIS-anchored GT of identical geometry, and COCO’s IoU computation is translation-invariant, but the two GT counts differ (6,980 tile annotations vs. 1,935 full-frame). A single target near a tile boundary appears in two overlapping tiles and is counted twice in the tile GT — so tile AP rewards high recall on near-boundary targets relative to full-frame AP. We flag this asymmetry as a caveat rather than a confound: the rank ordering within each scope is internally consistent.

Why the classical literature reports much higher numbers.

The PPINet paper [6] and Marine-Faster R-CNN [7] both evaluate against the DBSCAN padded GT (33×33 boxes) that we replaced. Uniform-size padded GT is much easier for any detector to match at $IoU = 0.50$ because box extent is decoupled from the actual radar return. Our numbers report **tight-box** IoU against AIS-verified targets and are therefore strictly lower than published results — but also more predictive of operational detection quality.

Limitations. Our Wavelet-SWT implementation emits 5 k predictions of mean size 44×47 px on a GT of median 15×15 — the method produces target-like blobs at the wrong scale and our per-component scoring doesn’t recover precision. A proper Wavelet-SWT evaluation would need per-scale bbox extraction and dedicated NMS. U-Net training did not complete in time for this draft.

CONCLUSION

We presented the first controlled cross-paradigm benchmark of classical and deep-learning object detection on X-band marine radar PPI imagery after fixing two systematic issues in the public DLR release (a 17-px polar-origin bias on all three sub-datasets and a $2\times$ range-scale error on MANV) and replacing the default DBSCAN padded GT with a hybrid AIS-anchored / image-grounded construction. Under this cleaner evaluation, YOLOv11 leads at $AP@0.50 = 0.776 \pm 0.06$ over 3 seeds, with RT-DETR at 0.738 and Faster R-CNN at 0.545. Classical methods are competitive but below the learned detectors in aggregate; their relative ranking depends sharply on the radar’s range-resolution regime, with CA-CFAR dominating the close-range MANV capture at $AP = 0.785$ and morphological / Otsu leading the long-range DAAN / DARC captures. All artefacts — code, configs, checkpoints, per-dataset calibration, hybrid-GT generator — are released to catalyse further research on radar-native deep architectures, self-supervised pretraining, and operational deployment of hybrid detection pipelines.

REFERENCES

- [1] H. M. Finn and R. S. Johnson, "Adaptive detection mode with threshold control," *RCA Review*, vol. 29, no. 9, 1968.
- [2] H. Rohling, "Radar CFAR thresholding in clutter and multiple target situations," *IEEE Trans. AES*, vol. AES-19, no. 4, 1983.
- [3] V. Duk et al., "Target detection in sea-clutter using stationary wavelet transforms," *IEEE Trans. AES*, vol. 53, no. 3, 2017.
- [4] Q. Li et al., "A speed-up seed point Otsu method for ship detection," *Proc. IEEE IGARSS*, 2016.
- [5] Y. Xu et al., "Adaptive clustering-based marine radar sea-clutter normalisation," *J. Sensors*, 2021.
- [6] X. Chen et al., "Automatic detection of scanning radar images based on Radar PPI-net," *Remote Sensing*, vol. 13, 2021.
- [7] X. Chen et al., "Marine-Faster R-CNN for navigation radar PPI images," *Front. Inf. Technol. Electron. Eng.*, vol. 23, 2022.
- [8] M. Heuer et al., "Marine radar object detection using SAHI-enhanced YOLOv11-OBb," *Sensors*, vol. 26, 2025.
- [9] J. Yee, "Onet: Twin U-Net for semantic segmentation in radar images," *IEEE Trans. Image Process.*, 2025.
- [10] L. Zhang et al., "CCDN-DETR for multi-class SAR object detection," *Sensors*, vol. 24, 2024.
- [11] D. Bolya et al., "TIDE: A general toolbox for identifying object- detection errors," *Proc. ECCV*, 2020.
- [12] X. Zhang and X. Li, "Survey on deep-learning-based marine object detection," *J. Adv. Transp.*, 2021.